

Module Description

EI10007: Principles of Information Engineering

Department of Computer Engineering

Module level: not specified	Language: English	Module duration: one semester	Occurrence: winter semester
Credits*: 5	Total number of hours: 150	Self-study hours: 105	Contact hours: 45

* The number of credits can vary depending on the corresponding SPO version. The valid number is always indicated on the Transcript of Records or the Performance Record.

Description of achievement and assessment methods:

The module examination is based on a written exam (75 minutes) which contains questions to assess the students' knowledge about the technical systems and their theoretical background, short mathematical problems to assess the students' mastering of the practiced mathematical concepts, and conceptual questions (e.g., about design principles or fundamental limitations) to assess the further intended learning outcomes. Up to 20% of the examination can be conducted in the form of multiple choice questions.

Possibility of re-taking:

In the next semester: Yes
At the end of the semester: No

(Recommended) requirements:

The following module should be successfully completed prior to participation: MA9711 Mathematics in Natural and Economic Science 1.

The following module is recommended to be attended in parallel (if not already attended earlier): MA9712 Statistics for BWL.

Contents:

* Fundamentals:

- Elements of Stochastic Modeling and Analysis
- Signals (analog/digital, deterministic/stochastic, real/complex)
- The Frequency Domain (Fourier transform, spectrum and bandwidth, sampling theorem)
- Information Theory (fundamentals, source coding, channel coding, channel capacity)

* Information Transmission and Storage Systems:

- Elements of Data Transmission (transmission chain, filtering, modulation, detection)
- Communication Systems (real systems compared to theory, channel models, performance criteria, comparison to data storage, current trends)
- Communication Networks (network structures, interference, broadcast and multiple access, multihop and relaying, abstraction layers, network planning)

* Elements of Information Processing

- Data Processing Devices (abstraction layers, real systems compared to theory, digital processing, algorithms and complexity)
- Data Acquisition (sampling and quantization, information and noise modeling)

Study goals:

After attending the module, the students:

- can describe the main principles of operation of information transmission systems and networks as well as of data processing devices
- are familiar with fundamental design principles of such systems and understand why existing systems are designed the way they are
- have an overview of the underlying physical and mathematical principles and can distinguish fundamental limitations from technological constraints
- have learned to take an engineering perspective on information transmission and processing tasks (e.g., by structuring a system into building blocks and abstraction layers)
- know the main mathematical methods relevant for this field of engineering and are able to apply a selection of these methods to example problems

Teaching and learning methods:

The module is designed for non-engineering students (in particular students in Management and Technology) who aim at understanding the fundamental principles and concepts of modern information transmission and processing. It consists of lectures, exercise courses, and self-study.

In the lectures, both theoretical backgrounds and technical implementations are introduced and discussed. Mathematical concepts are introduced and explained as far as it is necessary for understanding the technical systems. The relevance of each of the considered topics is motivated by, e.g., press articles, teaser questions, or examples from daily life, and an additional reflexion at the end of each topic unit aims at conveying the engineering perspective on the considered problems and systems. New concepts are presented in a teacher-centered style and discussed in an interactive manner.

The aim of the exercise courses is to repeatedly practice the application of the mathematical concepts as well as the ability to answer conceptual questions about the subject. The exercise courses are held in a student-centered way, and problem sheets are provided.

Throughout the semester, short reading assignments may be given to the students, e.g., as an introduction to a new topic. In addition, the students are expected to recapitulate the lecture contents and to individually practice the exercises.

Media formats:

- Slide Presentations
- Blackboard (e.g., for mathematical details)
- Supporting documents (e.g., news articles, scientific publications) as downloads (reading assignments)
- Problem sheets as downloads

Literature:

Since text books on the topics covered in the course are designed for a different target audience (engineering students), it is recommended to rely on the learning material provided during the course instead of on text books. However, students who are interested in more mathematical and technical details may refer to the corresponding sections in, e.g., the following text books.

- Bruce Hajek, "Random Processes for Engineers," Cambridge University Press, 2015.
- John G. Proakis and Dimitris G. Manolakis: "Digital Signal Processing. Principles, Algorithms, and Applications," Pearson Prentice Hall, 4th ed., 2007.
- Thomas M. Cover and Joy A. Thomas, "Elements of Information Theory," 2nd ed. Hoboken, NJ: Wiley- Interscience, 2006.
- Andrea Goldsmith, "Wireless Communications," Cambridge University Press, 2005.

- David A. Patterson and John L. Hennessy, "Computer Organization and Design. The Hardware / Software Interface," Elsevier/Morgan Kaufman, 5th ed., 2014.

Responsible for the module:

Utschick, Wolfgang; Prof. Dr.-Ing.: utschick@tum.de

Courses (Type, SH) Lecturer:

0000005872 Principles of Information Engineering (3SWS VI, WS 2024/25)

Gründinger A, Utschick W

For further information about this module and its allocation to the curriculum see:

<https://campus.tum.de/tumonline/wbModHb.wbShowMHBReadOnly?pKnotenNr=1490252>

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